# INFOSYS MOCK TEST 1

Easy to Moderate

## Rules and Instructions

- Attempt all questions within the suggested time.
- Do not use calculators, search engines, notes or AI assistance.
- Select one correct answer for each MCQ.
- No negative marking is assumed in this practice test.
- For coding questions, write complete executable solutions.
- Explain your approach and complexity during interview rounds.
- These are practice questions based on common fresher recruitment topics, not an official Infosys question paper.

## ROUND 1: ONLINE ASSESSMENT

### Section A: Logical Reasoning

**10 Questions | Suggested Time: 15 Minutes**

1. If CAT is coded as DBU, how is DOG coded?
   A. EPH  B. EOG
   C. FPH  D. DPH

2. Find the next number: 3, 6, 12, 24, ...
   A. 36  B. 42
   C. 48  D. 54

3. Choose the odd one out.
   A. Square  B. Triangle
   C. Circle  D. Cube

4. A is taller than B, and B is taller than C. Who is shortest?
   A. A  B. B
   C. C  D. Cannot say

5. A person walks 5 km north, 3 km east, then 5 km south. Where are they from the starting point?
   A. 3 km east  B. 3 km west
   C. 5 km north  D. At the starting point

6. Complete the series: AZ, BY, CX, DW, ...
   A. EV  B. FU
   C. EX  D. EW

7. If today is Wednesday, what day will it be after 10 days?
   A. Friday  B. Saturday
   C. Sunday  D. Monday

8. Ravi is 7th from the left and 9th from the right. How many people are in the row?
   A. 14  B. 15
   C. 16  D. 17

9. All roses are flowers. Some flowers fade quickly. Which conclusion definitely follows?
   A. All roses fade quickly  B. Some roses fade quickly
   C. All roses are flowers  D. No flower is a rose

10. Find the missing term: 2, 5, 10, 17, 26, ...
   A. 35  B. 36
   C. 37  D. 38

## Section B: Mathematical Ability

**10 Questions | Suggested Time: 25 Minutes**

1. A product costing Rs. 800 is sold for Rs. 920. Find the profit percentage.
   A. 10%  B. 12%
   C. 15%  D. 20%

2. A train travels 180 km in 3 hours. Find its average speed.
   A. 50 km/h  B. 55 km/h
   C. 60 km/h  D. 65 km/h

3. What is 25% of 360?
   A. 80  B. 85
   C. 90  D. 95

4. The average of 12, 15, 18 and 23 is:
   A. 16  B. 17
   C. 18  D. 19

5. A completes a task in 10 days and B in 15 days. Together they complete it in:
   A. 5 days  B. 6 days
   C. 7 days  D. 8 days

6. The ratio of boys to girls is 3:2. If there are 40 students, how many are girls?
   A. 12  B. 14
   C. 16  D. 18

7. Find the simple interest on Rs. 5,000 at 8% per annum for 2 years.
   A. Rs. 600  B. Rs. 700
   C. Rs. 800  D. Rs. 900

8. A number is increased by 20% and becomes 240. The original number is:
   A. 180  B. 190
   C. 200  D. 220

9. A bag contains 4 red and 6 blue balls. Probability of selecting a red ball is:
   A. 1/5  B. 2/5
   C. 3/5  D. 4/5

10. The perimeter of a square is 48 cm. Its area is:
   A. 100 $cm^2$  B. 121 $cm^2$
   C. 144 $cm^2$  D. 169 $cm^2$

## Section C: Verbal Ability

**10 Questions | Suggested Time: 12 Minutes**

1. Choose the synonym of Brief.
   A. Lengthy  B. Concise
   C. Unclear  D. Distant

2. Choose the antonym of Expand.
   A. Increase  B. Enlarge
   C. Contract  D. Extend

3. Choose the correct sentence.
   A. She do not like tea.  B. She does not likes tea.
   C. She does not like tea.  D. She not like tea.

4. He has been working here _____ 2022.
   A. for  B. since
   C. from  D. by

5. Choose the correctly spelt word.
   A. Accomodation  B. Acommodation
   C. Accommodation  D. Acomodation

6. The meeting starts _____ 9 a.m.
   A. in  B. on
   C. at  D. by

7. Choose the passive form: The team completed the task.
   A. The task completed the team.  B. The task was completed by the team.
   C. The task is completing by the team.  D. The team was completed by the task.

8. She is _____ honest person.
   A. a  B. an
   C. the  D. no article

9. Choose the word closest in meaning to Reliable.
   A. Trustworthy  B. Careless
   C. Temporary  D. Doubtful

10. Identify the error: Each of the students have a laptop.
    A. Each  B. students
    C. have  D. laptop

## Section D: Pseudocode and Output Tracing

**5 Questions | Suggested Time: 10 Minutes**

1. What is printed?
   `x=2; for i=1 to 3:  x=x*i; print(x)`
   A. 6  B. 8
   C. 12  D. 16

2. What is printed?
   `a=5; b=2; print(a mod b + a/b)`
   Assume integer division.
   A. 1  B. 2
   C. 3  D. 4

3. How many times does the loop execute?
   `i=1; while i<10:  i=i+3`
   A. 2  B. 3
   C. 4  D. 5

4. What is printed?
   `arr=[2,4,6]; print(arr[0]+arr[2])`

Assume zero-based indexing.
A. 6 B. 8
C. 10 D. 12

5. What is printed?
```
n=4; if n>5 print("A") else print("B")
```
A. A B. B
C. AB D. No output

### Section E: Puzzle Solving

**4 Questions | Suggested Time: 8 Minutes**

1. Three boxes are labelled Apples, Oranges and Mixed. Every label is wrong. You may take one fruit from one box. From which labelled box should you pick to identify all boxes? Explain.

2. Four friends P, Q, R and S sit in a row. P is left of Q, R is right of Q, and S is left of P. Write their order.

3. A clock shows 3:00. Find the smaller angle between the hour and minute hands.

4. You have 8 identical-looking balls; one is heavier. What is the minimum number of balance-scale weighings needed to guarantee finding it? Explain.

### Section F: English Writing

**Suggested Time: 10 Minutes**
Write 150–200 words on:
*How technology can improve the quality of education.*
Include an introduction, two supporting points, one example and a conclusion.

## ROUND 2: CODING ASSESSMENT

**Suggested Time: 45 Minutes**

1. **Second Largest Distinct Element**

   Given an integer array, print its second-largest distinct value. If it does not exist, print -1.

   Example:

   Input: `[4,9,2,9,7]`

   Output: `7`

   Expected approach: O(n).

2. **Character Frequency**

   Given a string, print the frequency of each character in order of first appearance.

   Example:

   Input: `banana`

   Output: `b:1, a:3, n:2`

   State time and space complexity.

## ROUND 3: TECHNICAL INTERVIEW

1. Introduce yourself and summarise your technical interests.
2. Explain the four pillars of object-oriented programming with examples.
3. What is the difference between an interface and an abstract class in Java?
4. Differentiate method overloading and method overriding.
5. What is the difference between == and equals() in Java?
6. Explain the difference between an array and a linked list.
7. What is a stack? Give two real-world applications.
8. Explain binary search and state its time complexity.
9. What is the difference between a primary key and a foreign key?
10. Explain database normalization and why it is useful.
11. Write an SQL query to display the number of employees in each department.
12. What are the ACID properties of a database transaction?
13. What is a process? How is it different from a thread?
14. What is deadlock? Name the conditions required for deadlock.
15. Explain the TCP three-way handshake.
16. What is the difference between HTTP and HTTPS?
17. Walk through one project on your resume: problem, architecture, contribution, challenge and outcome.
18. How did you test your project? Describe one bug you found and fixed.

## ROUND 4: HR AND BEHAVIOURAL INTERVIEW

1. Tell me about yourself in 60–90 seconds.
2. Why do you want to join Infosys?
3. What do you know about Infosys and the work it does?
4. What are your strengths? Support each with an example.
5. What is one weakness you are actively improving?
6. Describe a time you disagreed with a teammate. How did you resolve it?
7. Tell me about a failure or mistake and what you learned.
8. Are you comfortable relocating or working from another office location?
9. How do you manage deadlines when several tasks are pending?
10. Do you have any questions for the interviewer?

# ANSWER KEY

**Logical Reasoning:**
1-A, 2-C, 3-D, 4-C, 5-A, 6-A, 7-B, 8-B, 9-C, 10-C.
**Mathematical Ability:**
1-C, 2-C, 3-C, 4-B, 5-B, 6-C, 7-C, 8-C, 9-B, 10-C.
**Verbal Ability:**
1-B, 2-C, 3-C, 4-B, 5-C, 6-C, 7-B, 8-B, 9-A, 10-C.
**Pseudocode:**
1-C, 2-C, 3-B, 4-C, 5-B.
**Puzzle Guidance:**

1. Pick from the box labelled Mixed. It cannot be mixed, so the fruit identifies that box and the other two follow by elimination.

2. S-P-Q-R.

3. 90°.

4. Two weighings: weigh 3 against 3. If balanced, weigh the remaining two. Otherwise weigh two balls from the heavier group of three to isolate the heavy ball.